Michael Wieck-Sosa

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Education

Carnegie Mellon University | PhD in Statistics | Advisors: Aaditya Ramdas and Cosma Shalizi *Sept. 2022–Present*

- GPA: 3.98/4.00 | Awards: DeGroot-Goel Fellowship (2026), awarded by department faculty to one PhD student per year
- Thesis topics: generative models and time series | Committee: Aaditya Ramdas, Cosma Shalizi, Yuting Wei, Wei Biao Wu

University of Illinois at Urbana-Champaign | MS in Statistics *May 2022*

- GPA: 3.95/4.00 | Awards: two years of assistantships (tuition waiver and stipend) | GRE: 170/170 Q, 163/170 V, 4.5/6.0 W

Fordham University | BS in Mathematics with minors in Computer Science and Economics *May 2020*

- GPA: 3.77/4.00 | Awards: *magna cum laude* | Research: deep learning-based tracking system for an endangered toad species

Programming Languages and Tools

- Python expert: extensive experience with PyTorch, NumPy, SciPy, pandas, scikit-learn, and statsmodels
- R expert: extensive experience with dplyr, Rcpp, xts, zoo, caret, mgcv, glmnet, parallel, and ggplot2
- C++ proficient: used in Data Structures and Algorithms courses
- SQL, Git, Bash expert: extensive experience with data management, version control, and Linux

Coursework

- **Statistics**: Advanced Statistical Theory, Mathematical Statistics, Time Series Analysis, Regression, Computational Statistics
- **Computer Science**: Algorithms, Data Structures, Theory of Computation, Operating Systems, Computer Architecture, Artificial Intelligence, Machine Learning, Data Mining for Listening to the Social Universe
- **Mathematics**: Stochastic Calculus, Measure-theoretic Probability, Functional Analysis, Measure Theory, Interacting Particle Systems, Geometric Flows, Differential Geometry, Lie Groupoids and Algebroids, Topology, Abstract Algebra, Numerical Analysis, Numerical Linear Algebra, Real Analysis, Differential Equations, Linear Algebra, Mathematical Modeling

Research Assistantships and Internships

Carnegie Mellon University | NSF Grant (PI: Cosma Shalizi) | Research Assistant *May 2024–Present*

- Developed theory, methods, and software for generative modeling and mechanistic modeling for high-dimensional sequences
- Trained transformers, recurrent neural networks, and other neural network architectures on GPUs using PyTorch

National Center for Supercomputing Applications | Great Lakes to Gulf Project | Research Assistant *Sept. 2020–May 2022*

- Built confidence bands for chemical concentrations and fluxes in water over time at 1,000+ sites via parallel computing in R

MIT Lincoln Laboratory | Group 38 (Sensor Technology) | Research Intern *May 2021–July 2021*

- Tracked objects in outer space and conducted simulation studies to compare different tracking methods using MATLAB

University of Illinois at Urbana-Champaign | Computer Science Department | Research Assistant *Jan. 2021–May 2021*

- Modeled dynamics of misinformation spreading across multiple social media platforms using Hawkes processes with Python

Papers

- MWS, Michel Haddad, and Aaditya Ramdas. "The dynamic generalized covariance measure for conditional independence testing with nonstationary time series". Accepted in the Journal of Business & Economic Statistics. arXiv. Slides. Code.
- MWS and Cosma Shalizi. "Estimating dynamic models by matching random features". Submitted to the Journal of the Royal Statistical Society, Series B. arXiv. Code.
- MWS and Cosma Shalizi. "Generative sequence modeling for infinite memory processes via predictive states". Submitted to ICLR 2027. arXiv.
- MWS and Cosma Shalizi. "Compressed simulation-based inference: neural posterior estimation for massive data via low-dimensional sketches". Available upon request.
- MWS and Cosma Shalizi. "Dynamic models with p parameters are identified by 2p+1 random features". Major revision at Physical Review E. arXiv. Code.

MANUSCRIPTS IN PREPARATION

- MWS, Aaditya Ramdas, and Cosma Shalizi. "Distribution-uniform anytime-valid inference for time series".
- MWS, Aaditya Ramdas, and Cosma Shalizi. "Anytime-valid invariance testing for time series".
- MWS, Aaditya Ramdas, and Cosma Shalizi. "Model-agnostic simulation-based inference".
- MWS, Aaditya Ramdas, and Cosma Shalizi. "Pretraining large predictive state space models".

PROFESSIONAL ACTIVITIES

- Paper reviewer | Journal of the Royal Statistical Society, Series B (Statistical Methodology) *2026–Present*
- Committee chair | CMU Statistics PhD student committee for organizing industry and academia career events *2024–2025*

CONFERENCES AND WORKSHOPS

- Accepted participant | 2027 Workshop on Information-Theoretic Methods for Learning Dynamics | Program: Image-Based Scientific ML for Theories of Biological Dynamics Across Scales | National Institute for Theory and Mathematics in Biology
- Invited talk on conditional independence testing | 2026 Workshop on Sequential Methods (IWSM) | American University
- Contributed poster on simulation-based inference | 2025 Workshop on Neural SBI | Carnegie Mellon University
- Contributed poster on conditional independence testing | 2024 NBER-NSF Time Series Conference | Univ. of Pennsylvania